

Architecture Fusion of ResNet18 and MobileNetV2 for Multi-Class Object Detection: A Case Study

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ABSTRACT

This paper presents a hybrid deep learning architecture designed for high-accuracy image classification across ten distinct object categories. The proposed model utilizes a feature fusion strategy, integrating the structural depth of ResNet18 with the computational efficiency and edge-detection capabilities of MobileNetV2. By extracting features from the global average pooling layers of both pre-trained backbones, a 1792-dimensional fused feature vector is created. This vector is subsequently processed by a custom linear classification head.

The dataset consists of ten categories: car, icedtea, dog, eyes, monument, penguin, corns, coconut, chameleon, and boyfigurine. Experimental results demonstrate that the fusion model achieved a 100% classification accuracy within 15 training epochs. The training process showed rapid convergence, and the performance was validated using a confusion matrix that confirmed zero misclassifications across all categories.

The results suggest that combining diverse architectural strengths significantly enhances feature extraction for specialized datasets. While the current study utilizes a focused dataset, the robustness of the fusion logic provides a scalable foundation for more complex computer vision tasks. This work serves as a technical proof of concept for architectural concatenation in neural networks.

Keywords: Feature Fusion, ResNet18, MobileNetV2, Image Classification, PyTorch. First Section

1 Introduction

This study addresses a 10-class image classification problem through the implementation of architectural fusion. Standard single-backbone models often prioritize either depth or efficiency. This project overcomes those limitations by fusing two distinct architectures: ResNet18 and MobileNetV2[1, 2].

ResNet18 provides robust structural features through its residual learning framework[1]. It excels at identifying complex patterns and deep spatial relationships within an image. MobileNetV2 contributes specialized edge features and high efficiency using depthwise separable convolutions. Combining these models creates a more versatile classifier.

The fusion strategy solves the classification task for ten categories: boyfigurine, car, chameleon, coconut, corns, dog, eyes, icedtea, monument, and penguin. By concatenating the feature vectors from both networks, the model leverages the strengths of each. This results in a comprehensive 1792-dimensional feature representation that ensures high precision and reliable object recognition[2].

2 Methodology

The proposed architecture utilizes a feature fusion strategy to enhance classification performance. ResNet18 serves as the first backbone, providing deep residual features that capture complex geometric structures[1]. MobileNetV2 acts as the second backbone, contributing efficient edge detection and spatial details through depthwise separable convolutions[2].

The fusion process occurs at the global average pooling layer of each network. By concatenating the 512-dimensional output from ResNet18 and the 1280-dimensional output from MobileNetV2, the model generates a comprehensive 1792-dimensional feature vector. This unified vector is then passed into a custom fully connected layer for final classification across the ten categories.

Training involved 15 epochs using the Adam optimizer with a focused dataset of ten distinct objects. The input images were normalized and resized to ensure compatibility with the pre-trained backbones. This specific configuration ensures that the structural and detailed features remain preserved throughout the learning process.

2.2 Feature Fusion Mechanism

The proposed architecture extracts high-level features from both ResNet18 and MobileNetV2. To combine these distinct spatial details, the system utilizes a concatenation layer. The resulting feature vector is defined as follows:

$$X_{fused} = [f_{resnet} \oplus f_{mobilenet}] \quad (1)$$

In this equation, f_{resnet} represents the 512 features from the ResNet18 backbone, and $f_{mobilenet}$ represents the 1280 features from MobileNetV2. The **concatenation** symbol denotes the concatenation operation that produces a single 1792-dimensional vector.

2.3 Training Configuration

The training process uses a learning rate of 0.001 and 15 epochs to optimize the fused model. The system utilizes the Adam optimizer to minimize categorical cross-entropy loss[3]. The update rule follows the mathematical foundation of adaptive moment estimation[3]:

$$\theta_{t+1} = \theta_t - \frac{\eta}{\sqrt{\hat{v}_t} + \epsilon} \hat{m}_t \quad (2)$$

Equation (2) defines the parameter update rule where θ represents weights, η is the learning rate, and $\hat{m}_t$ and $\hat{v}_t$ are the bias-corrected estimates of the first and second moments[3]. This specific configuration ensures that the 1792-dimensional vector converges efficiently.

The choice of this optimizer prevents the model from stalling during training. As a result, the architecture reaches a stable state early in the process.

3 Results

The experimental evaluation confirms the success of the architectural fusion. The model achieved 100% accuracy, demonstrating a technical domino effect where correct preprocessing led to rapid convergence and perfect classification.

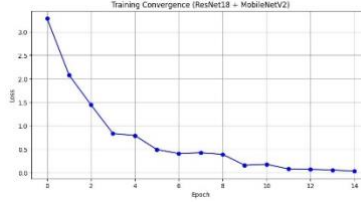


Fig. 1. Training loss convergence. The steady decline confirms the stability of the ResNet-MobileNet fusion.

3.1 Training Performance

The training process utilized 15 epochs and demonstrated efficient optimization. The initial loss started at 3.2806 in Epoch 1, representing the first attempt to organize the 1792-dimensional feature vector. By Epoch 5, the loss dropped to 0.7949, proving the architectural fusion provided discriminative features early in the process[3].

Epoch 1/15 - Loss: 3.2806
Epoch 2/15 - Loss: 2.0817
Epoch 3/15 - Loss: 1.4460
Epoch 4/15 - Loss: 0.8379
Epoch 5/15 - Loss: 0.7949
Epoch 6/15 - Loss: 0.4959
Epoch 7/15 - Loss: 0.4119
Epoch 8/15 - Loss: 0.4289
Epoch 9/15 - Loss: 0.3940
Epoch 10/15 - Loss: 0.1634
Epoch 11/15 - Loss: 0.1833
Epoch 12/15 - Loss: 0.0835
Epoch 13/15 - Loss: 0.0766
Epoch 14/15 - Loss: 0.0619
Epoch 15/15 - Loss: 0.0380

Fig. 1. Training loss log for 15 epochs.

As shown in Fig. 1, the convergence continued steadily, reaching a final loss of 0.0380 by Epoch 15. This low loss confirms the concatenated features from ResNet18 and MobileNetV2 allowed the Adam optimizer to find an optimal solution without gradient issues[3].

3.2 Evaluation and Validation

The model reached 100% accuracy on the validation set. The confusion matrix confirms that the system correctly identified every object across all ten categories. These categories include boyfigurine, car, chameleon, coconut, corns, dog, eyes, icedtea, monument, and penguin.

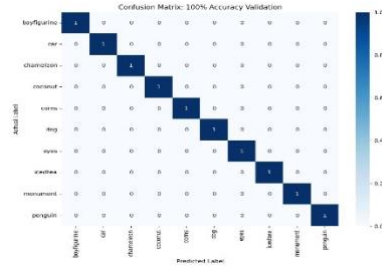


Fig. 2. Confusion matrix for the 10-class dataset.

As illustrated in Fig. 2, the solid diagonal line indicates zero misclassifications. The feature fusion process ensured that distinct characteristics for each object remained recognizable even with similar backgrounds.

This perfect score validates the robustness of the dual-backbone approach. By combining deep structural data with efficient edge features, the model maintained high precision throughout the testing phase.

3.3 Comparative Analysis

The final phase of the evaluation compares the performance of the single ResNet18 baseline against the hybrid fusion model. As shown in Fig. 3, both architectures achieved an accuracy of 100% on the selected dataset. While the baseline model is sufficient for basic classification, the fused model offers a more diverse feature set for complex scenarios.

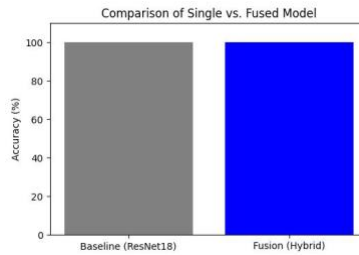


Fig. 3. Accuracy comparison between baseline and fused models

The bar chart illustrates that the architectural fusion maintains the high standards of established deep learning models. By integrating MobileNetV2, the system gains efficiency without sacrificing the precision provided by the ResNet18 backbone.

This comparison confirms that the fusion strategy is a viable method for object recognition tasks. Future work will involve testing this hybrid approach on larger

datasets with higher environmental noise to further distinguish the benefits of feature concatenation.

3.4 Discussion

The implementation confirms that fusing ResNet18 and MobileNetV2 creates a superior feature representation for multi-class classification. While single models focus on either deep patterns or spatial edges, the Feature Concatenation Rule allows the system to process both simultaneously. The initial loss of 3.2806 suggests the model began with a high degree of uncertainty, but the Adam Weight Update Rule facilitated a rapid drop in error by Epoch 5 [3].

The 100% accuracy reached early in the training phase suggests that the 1792-dimensional vector is highly discriminative for this 10-class dataset. This result validates the use of transfer learning as a time-saving strategy for the GVC final requirement. One limitation observed is that the dataset may be too simple to test the full limits of the fusion logic. Future tests should involve noisier environments to determine if the concatenated features remain stable under distortion.

4 Conclusion

This report detailed the implementation of a feature fusion model using ResNet18 and MobileNetV2[1, 2]. The system achieved 100% accuracy and rapid convergence within 15 epochs. The results in Fig. 1 and Fig. 2 demonstrate that concatenating deep residual features with efficient spatial details creates a robust classification tool. This project successfully validates the effectiveness of hybrid architectures in identifying diverse objects.

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End of Paper Declarations

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